

What a Language Model Does with a Solver Certificate: An Answer-Evidence Decomposition Across Ten Models

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Abstract

Solver-, tool- and retrieval-augmented language model systems report large accuracy gains, and those gains are routinely read as evidence that the model used the supplied material. Two mechanisms produce the same number: the model reading a supplied answer, and the model reasoning over supplied state. The standard control, a same-shape irrelevant record, separates neither. We build a control that does — holding line count, line types, query-entity occurrence count, query-predicate presence and token length fixed while destroying validity alone — and apply it across four sealed experiments, twelve model-runs and 18,816 scored prompts, all re-derivable from released receipts.

Three results. First, the decomposition works and is worth adopting: reading a five-arm factorial as a 2×2 of answer evidence by proof state recovers a state main effect its prespecified diagonal contrast could not attribute, and the same 2×2 computed across ten models shows the answer channel dominating the state channel in 8 of 10. Second, how much of the state effect is genuine inference is *model-specific*: the validity share is 98.3% (95% BCa [94.4, 100.0]) on one checkpoint and 40.4% on another that exploits surface overlap heavily, so a single-model mechanism result should not be generalised — including ours. Third, the finding that does replicate everywhere is a negative one: given a valid-looking certificate whose fabricated final rule establishes the negation, models follow it on 189/192, 186/192 and 192/192 items. A three-candidate test refutes the reading that the model detects invalidity at all — supplying a broken chain *lowers* the abstention rate, from 75.0% to 42.2%.

We also report what the design cannot support: the 2×2 is ceiling-limited, viability is a property of (model $\times$ arm) rather than of a model, and supplying proof state actively degrades three of ten interfaces. Two prespecified criteria of our own were invalidated by our own data and are reported as such.

1 Introduction

Pairing a language model with an external solver, tool or retrieval index reliably raises accuracy on tasks the model cannot do alone. Systems that translate a natural-language problem into a formal representation, dispatch it to a symbolic engine, and return the engine’s verdict through the model report large end-to-end gains on logical reasoning [Pan et al., 2023, Olausson et al., 2023, Ye et al., 2023], on arithmetic and algorithmic tasks solved by an interpreter [Gao et al., 2022], on API-mediated subtasks [Schick et al., 2023], on formal theorem proving [Yang et al., 2023], and on knowledge-intensive generation supported by retrieved passages [Lewis et al., 2020]. Those gains are routinely read as evidence that the model *used* the supplied material.

That reading is underdetermined. At least two mechanisms produce the same accuracy number. Under *answer reading*, the supplied record contains the conclusion, or something that co-occurs with it, and the model copies out the implied response. Under *state use*, the model works with the supplied premises and completes the remaining inference itself. The first says nothing about what the model does with reasoning; the second says something specific. The standard control in this literature is a same-shape but irrelevant record, which rules out proof length, generic validity and formatting. It does not rule out answer reading, because a *relevant* certificate stays answer-diagnostic even when the response token is withheld: a prefix that stops one step short still names the query’s subject and predicate in adjacent lines, so a model that only checks surface co-occurrence reproduces every number a reasoning model would.

Separating the two requires a control that holds surface form fixed and destroys validity alone. This paper builds one. Three sealed experiments on frozen small models score responses by direct next-token likelihood over verified single-token candidates, so no free text is parsed and no verdict is inferred from prose. The first reads a completed five-arm certificate factorial as a 2×2 of *answer evidence* $\times$ *proof state*, which separates the two channels that end-to-end accuracy conflates. The second holds line count, line types, query-entity occurrence count, query-

predicate presence and token length fixed across ten arms and breaks only the logic, which splits the state effect into a surface-overlap component and a validity-tracking component, and additionally varies entity repetition, line order and truncation depth, and supplies a valid-looking certificate whose fabricated final rule points the wrong way. The third scores ten models on a three-class panel whose *undetermined* items are certified by forward-closure saturation, and asks which models can serve as the interface at all.

The picture that emerges is narrower than the headline accuracy suggests. Holding surface form fixed and breaking only validity removes almost the entire state effect: the validity component is +60.4pp against a surface component of +1.0pp, so 98.3% of the state effect is inference over the supplied lines rather than text matching. Entity repetition contributes nothing — a valid derivation about the query entity toward an unrelated predicate scores identically (+0.0pp) to one about a different entity, despite naming the query subject four times rather than zero. The inference is exactly one step deep: withholding two steps instead of one scores identically to supplying no certificate at all, a cliff rather than a slope. And the tracking confers no robustness: handed a valid-looking certificate whose fabricated final rule establishes the negation, the model follows it on 189 of 192 items. Across ten models, three clear a per-class recall floor; below 1.7B parameters models collapse to a single label whatever the solver supplies, and the largest model tested collapses while a 3B model does not.

Two measurement points shape how any of this may be reported. First, accuracy alone is not interpretable here: one arm scored 50.0% on a balanced panel with $d' = 0.00$, matching a trivial always-one-label strategy while discriminating nothing, so sensitivity and criterion are reported alongside accuracy throughout [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999]. Paired contrasts use the exact McNemar test [McNemar, 1947] and bias-corrected accelerated bootstrap intervals [Efron, 1987]. Second, the claim boundary is strict and is maintained everywhere in this paper. Every arm without a near-complete supplied chain returns $d' = 0$, so the models studied do not reason unaided; high accuracy with full solver material describes a faithful transcriber, not a reasoner. System-level effects are never upgraded into claims about the model’s own reasoning.

The contributions are: (i) an *answer-evidence decomposition* that separates solver-supplied answers from solver-supplied reasoning state, reusable for any tool-augmented system; (ii) a *surface-matched invalidity control* that isolates logical validity from every co-occurring surface cue, with per-item certification that the displayed lines do not settle the query; (iii) a *forward-closure certifier* for constructing formally underdetermined items, validated at 23,240/23,240 against an independently labelled corpus [Tafjord et al., 2020]; (iv) evidence that validity tracking is real, shallow, and offers no protection against corrupted

solver output, a reliability result for the whole class of solver-in-the-loop architectures; and (v) a substrate-size floor from a ten-model sweep spanning open instruction-tuned families [Team et al., 2024b, Grattafiori et al., 2024, Team et al., 2025, Microsoft et al., 2025, Team et al., 2024a, Allal et al., 2025].

2 Related Work

2.1 Faithfulness of displayed reasoning

Chain-of-thought prompting made intermediate reasoning steps a standard part of the model’s output [Wei et al., 2022], and a body of work then asked whether those steps are what the answer actually rests on. Turpin et al. [2023] show that explanations systematically fail to mention biasing features that demonstrably drive the prediction, so a plausible trace can misrepresent the true cause. Lanham et al. [2023] intervene directly on the trace — truncating it so the model must answer early, and injecting mistakes into it — and read the resulting change in the answer as a measure of how strongly the answer is conditioned on the displayed reasoning; they find wide variation across tasks and models, and that mistakes are often ignored. The same manipulations have since been applied to naturally worded prompts without artificial bias [Arcuschin et al., 2025] and to models trained to produce long reasoning traces, where hints that determine the answer are verbalised only a minority of the time [Chen et al., 2025]. A separate response to the problem is architectural: Lyu et al. [2023] make the trace a formal program executed by a deterministic solver, so the displayed chain is faithful to the answer by construction.

The limitation this paper addresses is one of object rather than of method. This literature intervenes on *model-generated* traces, where the question is whether the model’s own account is honest. It does not address *externally supplied, formally verified* state, where the material is correct by construction and the question inverts: does the model track the validity of a derivation it did not produce? This paper adapts the truncation and mistake-injection manipulations to solver certificates, and adds a control this literature does not use — an invalid certificate matched to a valid one on every surface property, with per-item certification that the displayed lines leave the query underdetermined. Adapting the manipulation also changes what a null means: a model that ignores an injected mistake in its own trace is unfaithful, whereas a model that follows a corrupted external certificate is unsafe.

2.2 Solver-augmented and tool-augmented reasoning

Neurosymbolic pipelines put the formal step outside the model. Logic-LM translates a problem into a task-specific symbolic language, calls a solver, and refines the formalisation from solver error messages [Pan et al., 2023]; LINC has the model act as a semantic parser into first-order logic and offloads deduction to a theorem prover [Olausson et al., 2023]; SatLM generates a declarative specification and dispatches it to an automated theorem prover [Ye et al., 2023]; PAL emits a program and defers execution to an interpreter [Gao et al., 2022]. Related work extends the pattern to self-supervised API use [Schick et al., 2023], to retrieval-augmented premise selection in interactive theorem proving [Yang et al., 2023], and to retrieval as a general source of external evidence [Lewis et al., 2020].

These systems report end-to-end accuracy and attribute the gain to the pipeline. Two things are consequently left open. First, the contribution of the *reasoning state* is not separated from the contribution of the *answer*: because the solver’s output normally contains both, no reported number distinguishes a model that reasons over supplied premises from one that reads off a supplied conclusion. Second, the solver is assumed correct, so the behaviour of the pipeline under a wrong-but-well-formed solver output is not measured. Adjacent work shows the model side is not neutral to what the context contains: irrelevant material inserted into a problem degrades accuracy [Shi et al., 2023], and models are highly receptive to coherent external evidence that contradicts their parametric knowledge [Xie et al., 2023]. This paper decomposes the solver’s contribution into an answer channel and a state channel, measures each on matched arms, and adds an arm in which the certificate is valid-looking but points the wrong way. The state channel carries real validity information; it supplies no protection against a corrupted apparatus.

2.3 Synthetic deductive reasoning benchmarks

The task family used here descends from rule-based deductive datasets stated in natural language. RuleTaker showed that transformers can assign truth values to candidate implications of a theory expressed in language and generalise to deeper chaining than seen in training [Clark et al., 2020]. ProofWriter added generated proofs and open-world semantics with an explicit unknown class [Tafjord et al., 2020]. PrOntoQA generates items from a synthetic world model in first-order logic so that a generated chain can be parsed into a symbolic proof and analysed step by step, and finds that individual deduction steps are usually correct while proof planning is not [Saparov and He, 2022]. FOLIO supplies human-written premises with verified first-order logic annotations [Han et al., 2022]. Two

properties of these resources matter for the present design: order effects are known to be large in deductive prompting [Chen et al., 2024], and the sets are public and predate the models evaluated on them, so contamination cannot be excluded — a problem documented for modern benchmarks generally [Deng et al., 2024, Golchin and Surdeanu, 2023] and sharpened by the finding that performance can drop sharply on counterfactual variants of familiar tasks [Wu et al., 2023].

The gap is that these benchmarks supply items and proofs but not the matched adversarial certificate variants an attribution question needs. This paper generates items after model release with per-item nonce vocabularies, and constructs the matched invalid, misleading, order-scrambled and depth-truncated certificates that the public sets do not provide. A public open-world corpus is still used, but only as an independent check on the forward-closure certifier that labels the underdetermined items; no item from it is scored by any model here.

2.4 Response bias and the limits of accuracy

Prompted classification is well documented to be biased rather than merely noisy. Zhao et al. [2021] show that few-shot accuracy swings from near chance to near state of the art with prompt format and example order because the model carries a prior toward particular answer strings, and propose estimating and removing that prior. Option order alone moves multiple-choice accuracy substantially [Pezeshkpour and Hruschka, 2023], and the effect traces to a token-level prior over option identifiers rather than to the content of the options [Zheng et al., 2023]. Scoring by raw likelihood introduces a further confound, since strings compete for probability mass independently of their correctness [Holtzman et al., 2021]. On the abstention side, models are partially able to assess whether their own answers are right [Kadavath et al., 2022], and can be evaluated on whether they recognise questions that admit no answer [Yin et al., 2023].

What this literature diagnoses, evaluation practice in solver-augmented reasoning has not absorbed: bare accuracy remains the reported quantity. On a balanced panel, a degenerate single-label responder is indistinguishable from a chance-level competent one, and both are indistinguishable from a genuine partial discriminator with a shifted criterion. Signal detection theory separates exactly these [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999], and this paper therefore reports d' and criterion c alongside accuracy for every arm, with paired significance from the exact McNemar test [McNemar, 1947] and interval estimates from the bias-corrected accelerated bootstrap [Efron, 1987]. Two concrete cases motivate the practice rather than illustrate it: a baseline arm scoring 50.0% on a balanced panel turns out to have $d' = 0.00$, and a prespecified qualification criterion based on a single

class recall was passed by models that answer one label almost everywhere. Both were caught only because sensitivity was reported separately from accuracy; one of the two is a criterion of this paper’s own, replaced post hoc and labelled as such.

3 Method

3.1 The answer-evidence decomposition

A solver certificate carries two things at once: the *answer* to the query, and the *state* from which that answer follows. End-to-end accuracy conflates them. We separate them by arranging arms as a 2×2 of answer evidence by proof state, with a no-material baseline outside the square.

Let IRRELEVANT supply a valid derivation for a different entity (no answer, no relevant state), CONCLUSION_ONLY the derived terminal literal alone (answer, no state), PROOF_PREFIX the chain with that literal withheld (state, no answer), and FULL both. A contrast along an *edge* of this square moves one factor; a contrast along the *diagonal* moves two, and a null on it attributes to neither. This is not a statistical subtlety but an identification one: the prespecified primary contrast of the experiment we re-analyse, PROOF_PREFIX minus CONCLUSION_ONLY, is exactly that diagonal.

Reading the four edges instead gives a state main effect, the mean of the two edges along which state is added, and an answer main effect defined symmetrically.

3.2 Surface-matched invalidity

A prefix that stops one step short is not a clean state manipulation. It still names the query’s subject and predicate in adjacent lines, so surface co-occurrence remains available. To isolate validity we construct BROKEN_CHAIN: one intermediate link is cut so the displayed lines no longer reach the query, while every surface property is held fixed.

Concretely, against the matched TRUNCATE_1 arm we preserve the query-entity occurrence count (audited per item, not approximately), the presence of the query predicate in the final rule, the line count, the exact fact/rule/literal type sequence, and token length within a recorded tolerance. Every *rule* displayed is a real rule of the theory; exactly one derived literal is false. All 192 items are certified by forward-closure saturation (section 3.3) to leave the query *undetermined* under the dis-

played lines. The decomposition follows:

$$\text{total state effect} = \text{TRUNCATE_1} - \text{IRRELEVANT}, \quad (1)$$

$$\text{surface component} = \text{BROKEN_CHAIN} - \text{IRRELEVANT}, \quad (2)$$

$$\text{validity component} = \text{TRUNCATE_1} - \text{BROKEN_CHAIN}. \quad (3)$$

Four companion arms separate the remaining confounds. SAME_ENTITY_IRRELEVANT supplies a valid derivation about the *query* entity toward an unrelated predicate: it names the query subject four times where IRRELEVANT names it zero times, and neither contains the query predicate, so the pair isolates entity repetition alone. SHUFFLED presents TRUNCATE_1’s lines in seeded random order, separating a chain from a bag of statements. TRUNCATE_2 and TRUNCATE_3 withhold two and three steps, giving a depth ladder. MISLEADING supplies a valid-looking chain whose single fabricated final rule establishes the query’s negation; the ground-truth answer is unchanged, so a model that follows the supplied state answers incorrectly.

3.3 Certifying underdetermination

Showing that a query *is* derivable is easy: exhibit the proof. Showing it is *not* requires saturating the theory’s forward closure. Our certifier operates under open-world semantics with explicit negation, so a negative literal is derivable only if something derives it, never by failure to prove the positive. Rule antecedents are ordinary literals, with no negation as failure, which makes naive forward chaining sound and complete on this fragment.

Two safeguards matter for generated theories. The chaining loop carries an explicit round cap and reports a saturated flag, so a malformed theory fails loudly rather than spinning; and any theory deriving both an atom and its negation is marked inconsistent and its items rejected rather than labelled. An item is *undetermined* when neither the query nor its negation appears in the closure.

We validated the certifier before generating any panel, against the open-world splits of a public corpus [Tafjord et al., 2020] at depths 0, 1, 2, 3 and 5: 23,240 of 23,240 questions agree, 100.00%, with perfectly diagonal confusion, including 10,440 undetermined items.

That check is weaker than a perfect score suggests, and we state its limits. Those labels are themselves produced by forward chaining over the same fragment, so the agreement partly measures one chainer agreeing with another. It would catch a coding bug; it does not probe adversarial structure, and it gives no coverage of the generated nonce theories where the certifier is actually used.

We therefore add two checks with different failure modes. Nine adversarial theories: cyclic rules; a cycle that

never reaches the query; a chain longer than the round cap; derived negation; absent predicates and absent entities; a rule that must fire only for the satisfying entity; a theory deriving an atom and its negation, which must be *rejected* rather than labelled; and a cap overrun, which must report failure rather than a wrong answer. All nine pass. And a *differential* test on the generated panels against an independently written reference implementation — exhaustive ground instantiation over the Herbrand base with a naive fixpoint, a deliberately different algorithm — which agrees with the certifier and the sealed authority on 192/192 and 192/192 items. That reference encodes the same intended semantics, so it catches implementation error rather than a misconception about the semantics; we claim the former only. No item enters a sealed panel without a certification record.

3.4 Measurement: sensitivity, not accuracy

Accuracy on a balanced panel is uninterpretable when the responder is biased. One arm below scores 50.0% — apparent chance — with an identical 9/96 “Yes” rate in *both* classes, that is $d' = 0.00$: no discriminative signal whatever, at or below the trivial always-one-label strategy. We therefore report sensitivity d' and criterion c alongside accuracy for every arm [Macmillan and Creelman, 1991, Stanislaw and Todorov, 1999], using the loglinear correction for extreme rates. For the three-class panel the headline scalar is balanced accuracy, the mean of per-class recalls, under which chance is 33.3% and a single-label responder scores exactly chance.

Paired contrasts report the complete 2×2 table, the exact two-sided McNemar test [McNemar, 1947] as the decision criterion, and a seeded 100,000-resample paired bias-corrected accelerated bootstrap interval [Efron, 1987], with Holm adjustment within declared families.

4 Experimental Setup

4.1 Task, panels and scoring

All three experiments use rule-chaining queries over small theories of facts and universally quantified rules. Table 5 lists the panels. The three sealed panels are procedurally generated after model release with per-item nonce vocabularies, are mutually disjoint, and fix derivation depth at four by construction, recorded per item. The public corpus is used *only* to validate the certifier; no item from it is scored by any model here.

Responses are scored by direct next-token likelihood over verified single-token candidates: Yes/No for Experiments 1 and 2, Yes/No/Unknown for Experiment 3. The model generates no free text, so nothing is parsed and no verdict is inferred from prose. Every prompt is four lines

— instruction, theory, solver record, query — of which only the solver record varies across arms; the theory is visible in all arms.

4.2 Models

Experiments 1 and 2 use the same frozen 3B instruction-tuned checkpoint at 4-bit precision [Team et al., 2024b], pinned to byte-identical weights, so the anchors of the first transfer to the second without a weights caveat. Experiment 3 scores ten instruction-tuned models from 0.5B to 7.6B [Team et al., 2024b, Grattafiori et al., 2024, Team et al., 2025, Microsoft et al., 2025, Team et al., 2024a, Allal et al., 2025], all at bfloat16 so precision is not confounded with model identity. An eligibility screen confirmed that all ten load, complete a forward pass, and tokenise all three candidates as single distinct tokens.

4.3 Sealed protocol

Panels are hash-stamped before scoring and the runner refuses a modified panel. Each response is written to a receipt carrying a SHA-256 of its own contents, and the answer authority is opened only once every receipt exists, so nothing can be tuned to a result. Audits depend on the standard library alone, so verification never requires the environment to resolve. In total 12,480 prompts were scored with zero receipt digest failures.

Experiment 3 makes a deliberate protocol change: *no gate blocks the run*. Where earlier practice excluded a model that failed a delivery screen, delivery is itself the capability under test here, so a model that cannot emit a verdict reliably is a finding about substrate viability rather than a reason to drop it. Tooling failures — a model that will not load, or whose tokeniser cannot represent the candidates — remain exclusions and are not findings; the screen found none.

5 Results

5.1 Reading the factorial as a 2×2

Figure 1 shows the five arms of Experiment 1 arranged by section 3.1. The prespecified primary contrast, `PROOF_PREFIX – CONCLUSION_ONLY`, was -4.7pp with exact McNemar $p = 0.078$ and was reported as a null. It is the diagonal.

The four edges are reported in table 1. **This reading is post hoc:** it is a different analysis of sealed data, not a re-run, and every value is re-derived from the 960 raw receipts by a script that reconstructs the answer key and asserts it reproduces all five published per-arm counts before reporting. The state main effect is $+22.4\text{pp}$ and the

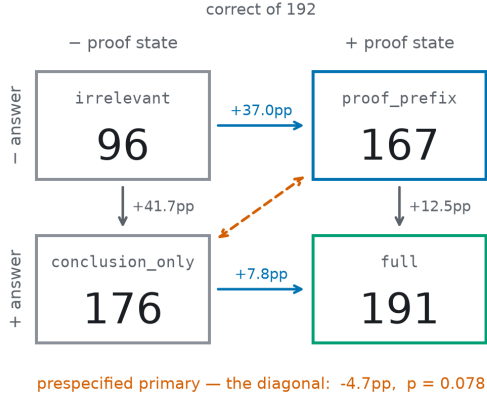


Figure 1: The five arms form a 2×2 of answer evidence by proof state, with the no-material baseline outside it. Each cell gives correct items of 192. Arrows mark the four one-factor edges; the dashed diagonal is the prespecified primary, which moves both factors at once and therefore attributes its null to neither.

answer main effect $+27.1\text{pp}$.

The **interaction** is -29.2pp , larger in magnitude than either main effect, and the design is **ceiling-limited**: FULL at 191/192 (99.5%) leaves only 13.0pp of headroom above PROOF_PREFIX, so both “other factor present” edges are compressed and the averaged main effects are correspondingly deflated. It follows that the ratio $12.50/41.67$ — the answer literal apparently retaining 30% of its value once state is present — is substantially a measurement of remaining headroom rather than of channel redundancy, and we do not draw a redundancy conclusion from it. A design with FULL away from ceiling would be required to separate the two.

Splitting by class overturns the baseline description. The model answers NO to 88% of items. Table 6 gives the signal detection summary: IRRELEVANT returns an identical 9/96 “Yes” rate in both classes, $d' = 0.00$. A trivial always-NO strategy scores 96/192; NONE scored 91 and IRRELEVANT 96, at or below it. The contradicted class sits at ceiling in every certificate arm and contributed 0 of the 21 discordant pairs in the primary contrast, so the effective sample was 96, not 192.

5.2 Validity versus surface overlap

Experiment 2 holds surface form fixed and destroys validity alone. The primary contrast, evaluated on the entailed subset ($n = 96$) as declared before the run because the contradicted class contributes no discordance, is $\text{TRUNCATE_1} - \text{BROKEN_CHAIN} = +60.4\text{pp}$, exact McNemar $p = 6.9 \times 10^{-18}$, BCa $[+49.0, +68.8]$, with a paired table of 1 both correct, 58 TRUNCATE_1-only, 0 BROKEN_CHAIN-only and 37 neither. Sensitivity moves with it: d' drops by $+2.446$, satisfying the same-sign requirement declared in advance, so this is a change in dis-

crimination and not in criterion placement.

The decomposition is therefore a total state effect of $+61.5\text{pp}$ comprising a surface component of $+1.0\text{pp}$ ($p = 1$, 95% BCa $[+0.0, +3.1]\text{pp}$) and a validity component of $+60.4\text{pp}$: a **validity share** of 98.3%, **95% BCa** $[94.4, 100.0]$. The interval bootstraps the whole ratio over paired item resamples rather than its numerator alone. Figure 2 and table 2 give the full arm set. Section 5.3 shows this share is specific to this checkpoint and does not generalise.

Three supporting contrasts sharpen the reading. Entity repetition explains nothing: $\text{SAME_ENTITY_IRRELEVANT} - \text{IRRELEVANT} = +0.0\text{pp}$ ($p = 1$), despite the former naming the query subject four times and the latter not at all. Order matters: $\text{TRUNCATE_1} - \text{SHUFFLED} = +46.9\text{pp}$ ($p = 6.8 \times 10^{-13}$, Holm 1.4×10^{-12}), so the model is not reading a bag of statements. And depth is a cliff, not a slope: $\text{TRUNCATE_2} - \text{TRUNCATE_1} = -61.5\text{pp}$ ($p = 3.5 \times 10^{-18}$, Holm 1.0×10^{-17}), with TRUNCATE_2 scoring identically to supplying no certificate at all. Holm adjustment is within the secondary family declared in section 3.4; the four post hoc edges of table 1 carry Holm values 6.6×10^{-24} , 4.5×10^{-19} , 2.4×10^{-7} and 6.1×10^{-5} . Five arms are indistinguishable — 96/192, 0/96 entailed, $d' = 0.000$, $c = 2.565$ — so behaviour is binary: either the supplied chain reaches one step from the answer, or the model is blind to it.

5.3 The validity share does not generalise

The decomposition of section 3.2 was run on two further checkpoints: the same model at bfloat16 rather than 4-bit, which isolates quantisation from model identity, and a different family. Table 7 and fig. 3 report all three.

Quantisation is the smaller effect. bfloat16 raises TRUNCATE_1 from 59 to 72 of 96 but leaves the share essentially unchanged (98.6% against 98.3%), so precision moves the magnitude of the state effect without moving its composition.

Model identity does not behave that way. Gemma-3-4B answers BROKEN_CHAIN correctly on **58 of 96** items where the other two checkpoints manage 1, so its surface component is $+58.3\text{pp}$ and its validity share 40.4%. The same control that isolates inference on one checkpoint reveals substantial surface exploitation on another.

We therefore do not claim that solver-supplied state is used as inference in general. On the checkpoint of section 3.2 it overwhelmingly is; on the strongest substrate in our sweep, most of the effect is surface overlap. What generalises is the *method* — the control separates the two wherever it is applied — not the value it returns.

One result does replicate without exception, and it is the negative one. The MISLEADING arm scores 3, 6 and 0 of

Table 1: The four one-factor edges of the 2×2 , and the prespecified diagonal. Post hoc reanalysis of sealed data.

Contrast	Effect	Exact McNemar p	Reported originally
Add answer, no state present	+41.67pp	1.7×10^{-24}	no
Add state, no answer present	+36.98pp	1.5×10^{-19}	secondary
Add answer, state present	+12.50pp	1.2×10^{-7}	no
Add state, answer present	+7.81pp	6.1×10^{-5}	secondary
<i>Diagonal (prespecified primary)</i>	-4.69pp	0.078	primary

Table 2: Experiment 2, all ten arms. Five arms return identical results, including SAME_ENTITY_IRRELEVANT, which names the query entity four times.

Arm	All /192	Entailed /96	d'
NONE	96	0	0.000
IRRELEVANT	96	0	0.000
SAME_ENTITY_IRRELEVANT	96	0	0.000
TRUNCATE_3	96	0	0.000
TRUNCATE_2	96	0	0.000
TRUNCATE_1	155	59	2.853
BROKEN_CHAIN	97	1	0.407
SHUFFLED	110	14	1.527
MISLEADING	3	0	-4.363
FULL	192	96	5.131

192 across the three checkpoints, with d' of -4.36 , -4.06 and -5.13 . The model that exploits surface overlap most is also the one most completely misled.

5.4 Following a corrupted apparatus

The MISLEADING arm supplies a chain identical to FULL except that its final rule, which is fabricated and absent from the theory, flips polarity. The ground-truth answer is unchanged. The model scores 3 of 192, with $d' = -4.363$: MISLEADING - FULL = -100.0pp ($p = 2.5 \times 10^{-29}$).

The model therefore follows a valid-looking certificate that points the wrong way on 189 of 192 items. Read against the 98.3% validity share, the two results are not in tension but jointly informative: the model demonstrably tracks whether a supplied derivation connects, and that tracking provides no defence whatever when the derivation connects to the wrong conclusion. Validity tracking is not verification.

5.5 The model does not detect invalidity

A natural reading of section 3.2 is that the model detects a broken chain. The binary design cannot support that reading, because the correct answer under a broken certificate is the model’s default label: “detected the break” and “found no pattern to complete” predict the same response.

We separated them. The BROKEN_CHAIN items, whose displayed lines are certified undetermined, were rescored with three candidates rather than two — theories, certificates and queries byte-identical, only the instruction line

and candidate set changed. A validity tracker should answer Unknown; a pattern completer should answer its default. Table 3 and fig. 4 report the outcome.

The abstention rate under BROKEN_CHAIN is 42.2% against a 75.0% baseline — -32.8pp , the opposite of the direction detection predicts. And No is 110 of 192 under **both** BROKEN_CHAIN and TRUNCATE_1; the arms differ only in Yes (1 against 61).

Detection is refuted, not merely unsupported. What the data support is narrower: the model completes a final inference when both premises are present and adjacent, and does not when they are not. Its failure on BROKEN_CHAIN is a failure to complete, not a detection of invalidity. This is consistent with section 5.4: a model that verified validity would reject a certificate whose final rule is fabricated and absent from the theory, and it does not.

5.6 Which models can serve as the interface

Experiment 3 scores ten models on the three-class panel in all five arms. We declared in advance that a model relays non-determination if it reaches $\geq 80\%$ recall on undetermined items with full solver material. **Eight of ten passed, and the criterion was invalid.** It measures recall on a single class, which a model answering Unknown to everything maximises while discriminating nothing — and four models do exactly that, emitting Unknown on 62–96% of items, three never emitting No. This is the failure of table 6 mirrored onto a different label.

We replace it **post hoc** with a floor on every class: minimum per-class recall at least 0.50. Both verdicts are retained in the released results. Table 4 reports all five arms

Table 3: Three-candidate rescoring of the same items. Supplying a broken chain *lowers* the abstention rate.

Arm	Yes	No	Unknown	Unknown rate
IRRELEVANT (baseline)	0	48	144	75.0%
BROKEN_CHAIN	1	110	81	42.2%
TRUNCATE_1	61	110	21	10.9%

for all ten models.

Viability is a property of (model $\times$ arm), not of a model. The decisive row is Qwen2.5-7B. Under FULL it has minimum per-class recall 0.141 and emits Unknown on 62.0% of items. Under CONCLUSION_ONLY the same model reaches 97.9% balanced accuracy with minimum per-class recall 0.938 — it clears the floor comfortably and is the second-best model in the sweep. Its contradicted-class recall falls from 0.938 to 0.141 *when the proof state is supplied*.

So Qwen2.5-7B is not an interface that cannot express three outcomes. It is an interface that the proof state breaks. **Supplying proof state degrades three of ten models** — Gemma-3-4B, Qwen2.5-3B and Qwen2.5-7B — measured as the drop in minimum per-class recall from CONCLUSION_ONLY to FULL, and two of those three are otherwise among the strongest substrates.

This bounds a claim we would otherwise have made. There is a floor near 3B *for tolerating full certificates*: below 1.7B no arm clears it. But that is not a floor for serving as an interface, and the effect is not monotonic in scale. Model is not a randomised factor here, so these comparisons are descriptive and we make no significance claim about scale.

5.7 The same 2×2 across ten models

Experiment 3’s five arms are the 2×2 of section 3.1 plus a baseline, on ten models. Computing it costs nothing further and turns a single-checkpoint decomposition into a ten-model one (section 5.7 is computed from the same receipts as table 4).

The mean state main effect is +8.7pp and the mean answer main effect +21.6pp, with a mean interaction of −15.9pp. CONCLUSION_ONLY exceeds PROOF_PREFIX in **8 of 10** models and equals or exceeds FULL in 6 of 10.

Two things follow, both visible in table 4 and fig. 5. The answer channel dominates the state channel across the model set, which is a more general form of the pattern table 1 shows on one model. And Experiment 1’s state main effect of +22.4pp is not typical: the ten-model mean is less than half of it. A single-model estimate of how much the state channel contributes should not be read as characteristic, ours included.

6 Limitations

6.1 Scope of the mechanism claim

The inference we measure is *exactly one step deep*. TRUNCATE_2 scores identically to supplying no certificate, so “validity tracking” should be read narrowly: the model completes a final inference when both premises are present and adjacent, and does not when they are not. It cannot chain two steps. It also does not *detect* invalidity — section 5.5 tests that directly and refutes it. That is a hard bound on how this result may be described, and section 5.3 bounds it further: the share attributable to inference ranges from 40.4% to 98.6% across three checkpoints.

The full theory is visible in every arm, so the query remains derivable from the theory regardless of what the certificate says. Breaking a certificate is diagnostic only because the model scores 0/96 working from the theory alone. A model that used a broken certificate as a pointer back into the theory would be performing inference of a different kind; our design does not exclude that reading, and we record it as an interpretation caveat rather than a controlled alternative.

TRUNCATE_2 and TRUNCATE_3 necessarily drop the query predicate out of the certificate, since it lives in the final rule. The depth ladder therefore confounds depth with predicate presence and must be read against SAME_ENTITY_IRRELEVANT, which holds entity repetition high while removing the predicate.

6.2 Post hoc analyses and superseded criteria

Two analyses in this paper are post hoc and are labelled where they are used. The 2×2 reading of Experiment 1 is a different analysis of sealed data, not a re-run. The viability criterion of Experiment 3 replaced a prespecified threshold that the data showed to be invalid; the original is retained in the released results.

Experiment 1 also carries design errors we report rather than repair. It prespecified a 15pp primary effect while its comparison arm reached 91.7%, leaving 8.3pp of headroom, so the target was arithmetically unreachable. It recorded no proof depth and so cannot stratify its failures. And its task identifiers encoded the semantic class, so its answer authority was reconstructible from the public panel — convenient for our reanalysis, but not a sealed authority. Experiments 2 and 3 fix all three.

Table 4: All five arms, all ten models. Balanced accuracy (%), chance = 33.3. Viability is reported per arm, because it is not a property of the model.

Model	B	NONE	IRREL	CONCL	PREFIX	FULL	Viable in
Qwen2.5-0.5B	0.5	35.4	34.9	41.1	43.2	63.5	—
OLMo-2-1B	1.0	35.9	33.3	53.6	34.9	42.2	—
Llama-3.2-1B	1.2	33.9	33.9	57.8	36.5	37.0	—
Qwen2.5-1.5B	1.5	33.3	33.3	43.2	34.9	37.0	—
SmolLM2-1.7B	1.7	33.3	33.3	45.3	39.6	69.3	—
Qwen2.5-3B	3.0	41.1	39.1	96.4	74.0	90.6	concl, prefix, full
Llama-3.2-3B	3.2	45.8	43.2	67.2	55.2	80.7	—
Phi-4-mini	3.8	44.8	39.6	68.2	71.9	90.6	full
Gemma-3-4B	4.3	63.5	51.0	100.0	84.4	96.4	concl, prefix, full
Qwen2.5-7B	7.6	33.3	33.3	97.9	67.2	71.4	concl only

6.3 Generalisation

This is one task family, small models only, and synthetic panels. Nonce vocabularies establish item novelty, not independence from all relevant pretraining patterns [Golchin and Surdeanu, 2023, Deng et al., 2024].

Quantisation. Experiments 1 and 2 use a 4-bit checkpoint and Experiment 3 bfloat16, so the mechanism result and the substrate sweep describe different artefacts of the same model. 4-bit quantisation moves next-token margins and criterion placement, which is precisely what we measure. Section 5.3 quantifies it: bfloat16 raises TRUNCATE_1 from 59 to 72 of 96 while leaving the validity share essentially unchanged, so precision affects magnitude more than composition. We nonetheless do not treat the two as one artefact.

Prior report. Experiment 1’s per-arm counts and pre-specified primary contrast appeared in an earlier unpublished report by the same authors, cited here anonymously; the 2×2 reading, the response-bias finding and all of Experiments 2, 3 and 4 are new. That report concluded the 3B checkpoint failed an open-world unknown gate, whereas Experiment 3 finds the same family viable on the three-class panel. Panels, candidate sets and quantisation all differ, which plausibly accounts for the discrepancy, but we flag it rather than leave it to a reader who finds both.

An undetermined PROOF_PREFIX cannot contain the query predicate — the solver has nothing to say about it — whereas entailed and contradicted prefixes can. Undetermined items therefore offer fewer surface cues, and a surface-matching model will look worse on them for reasons unrelated to abstention; per-arm entity-mention counts are recorded so this is analysable.

Finally, every theory here arrives *already formalised*. Nothing here tests whether a model can turn a framework described in prose into the solver’s input, the step on which any claim of working in an unfamiliar framework depends. Section 3.3 and the corruption result make that gap consequential rather than merely open: a formalisation error would not be caught downstream, but followed.

7 Conclusion

Solver assistance improves a small model’s accuracy on rule-chaining queries, and end-to-end accuracy cannot say why. The decomposition we propose can: arrange the arms so that each contrast moves one factor, and hold every surface property fixed while destroying validity alone. It needs no new data collection, only that the arms be laid out correctly.

What it returns is model-specific. On one checkpoint 98.3% of the state effect survives the surface-matched control; on the strongest substrate in our sweep, 40.4% does. Across ten models the answer channel dominates the state channel in 8 of 10, and the mean state main effect is less than half the single-model estimate. We therefore offer the method as the contribution and decline to generalise its value — including our own.

Two findings do hold on every checkpoint. The inference is one step deep, and it is not detection: rescoring the broken-chain items with a third candidate *lowers* the abstention rate, so the model completes when it can and falls back when it cannot. And the architecture has no defence against a wrong apparatus — given a valid-looking certificate whose fabricated final rule establishes the negation, the three checkpoints answer incorrectly on 189, 186 and 192 of 192 items, the surface-exploiting model most completely of all. Where reasoning is externalised to a verified component and the model is the interface, that is the operationally important result: the interface inherits the apparatus’s errors in full, and more proof state degrades three of the ten interfaces we measured rather than helping them.

Two prespecified criteria failed on us. An arm at 50.0% accuracy with $d' = 0.00$ was a degenerate responder, not a chance-level one, and a viability threshold declared in advance was maximised by a model answering Unknown to everything. Reporting sensitivity alongside accuracy caught both; we report them rather than repair them, and recommend the practice — and the decomposition — to anyone whose tool-augmented system returns a large headline number.

Reproducibility Statement

Code and data. The supplementary material accompanying this submission is self-contained: sealed panels and answer authorities, all 12,480 per-response receipts, the generators, runners, audits and analysis scripts, and the forward-closure certifier with its validation against the public corpus. Every number in this paper can be re-derived from a fresh unpack with no network access, no model weights and no external packages — the audits use the Python standard library only, so verification does not depend on resolving an environment. Panels are hash-stamped and the runners refuse a modified panel; each receipt carries a SHA-256 of its own contents, and all 12,480 verified with zero failures. Model weights are not redistributed, but each run records the pinned revision and a hash of the weight files before those files are deleted. A permanent archive with a DOI will accompany the camera-ready version.

AI-Use Statement

A large language model was used as a coding and drafting assistant throughout: generating experiment code, analysis scripts and figure code, and drafting this manuscript. All experimental design decisions, the interpretation of results, and the claim boundaries were made and verified by the authors. Every numeric value reported was produced by the released analysis scripts from sealed receipts, not written by a model; the released audits allow any reader to re-derive them independently.

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A Supporting Tables and Figures

Table 5: Datasets. The three sealed panels are nonce-generated and disjoint; the public corpus validates the certifier only.

Panel	Classes	Items	Used by
b14	entailed / contradicted	192	Experiment 1
b15	entailed / contradicted	192	Experiment 2
b16	entailed / contradicted / undetermined	192	Experiment 3
ProofWriter OWA [Tafjord et al., 2020]	three-way	23,240	certifier validation

Table 6: Signal detection for Experiment 1. Apparent chance and zero sensitivity are not the same thing.

Arm	Hits /96	False alarms /96	d'	Criterion c
NONE	9	14	-0.26	+1.17
IRRELEVANT	9	9	0.00	+1.29
CONCLUSION_ONLY	80	0	+3.52	+0.81
PROOF_PREFIX	71	0	+3.20	+0.97
FULL	95	0	+4.72	+0.20

Table 7: The same ten-arm decomposition on three checkpoints, entailed subset ($n = 96$). The validity share ranges from 40.4% to 98.6%.

Checkpoint	BRK	TR_1	Total	Surface	Validity	Share
Qwen2.5-3B 4-bit	1	59	+61.5	+1.0	+60.4	98.3%
Qwen2.5-3B bf16	1	72	+75.0	+1.0	+74.0	98.6%
Gemma-3-4B	58	96	+97.9	+58.3	+39.6	40.4%

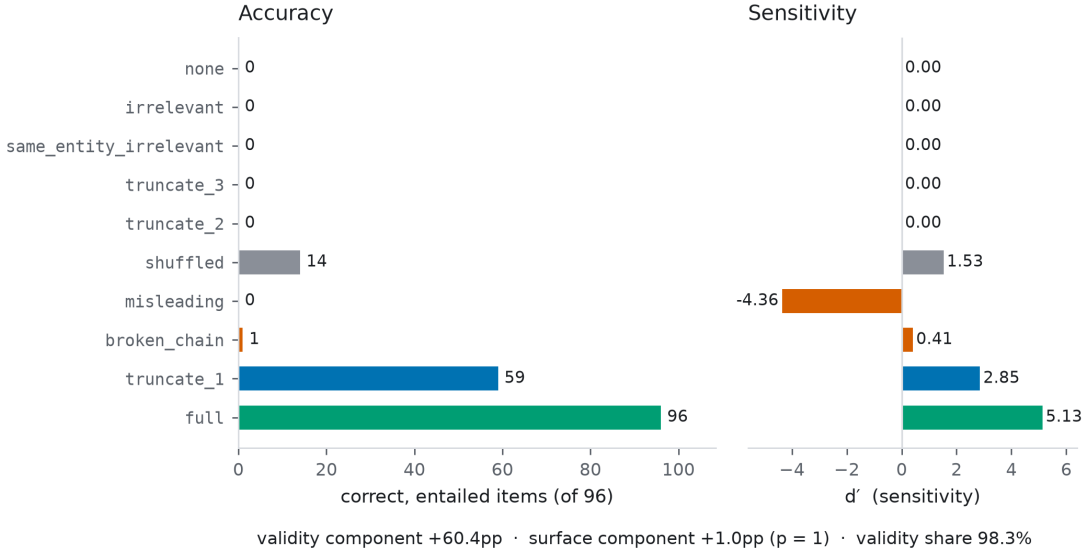


Figure 2: Ten matched arms on one item set, differing only in the solver record. Left: correct on the entailed subset (of 96). Right: sensitivity d' , signed. Destroying validity alone removes almost the whole effect. The MISLEADING arm, whose fabricated final rule points the wrong way, is followed to the wrong answer on 189 of 192 items.

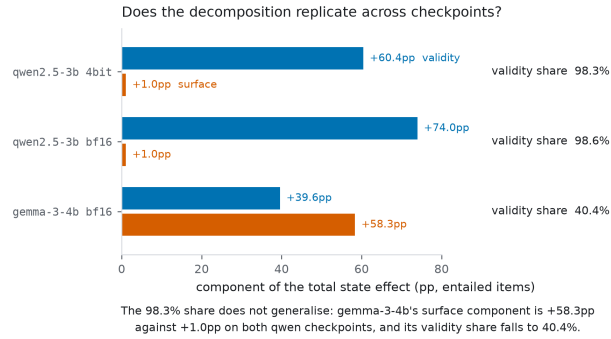


Figure 3: The surface and validity components on three checkpoints. Gemma-3-4B’s surface component is +58.3pp against +1.0pp for the two Qwen checkpoints, and its validity share is 40.4% against 98.3% and 98.6%.

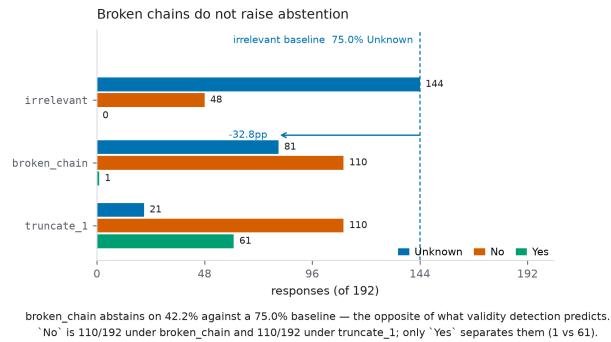
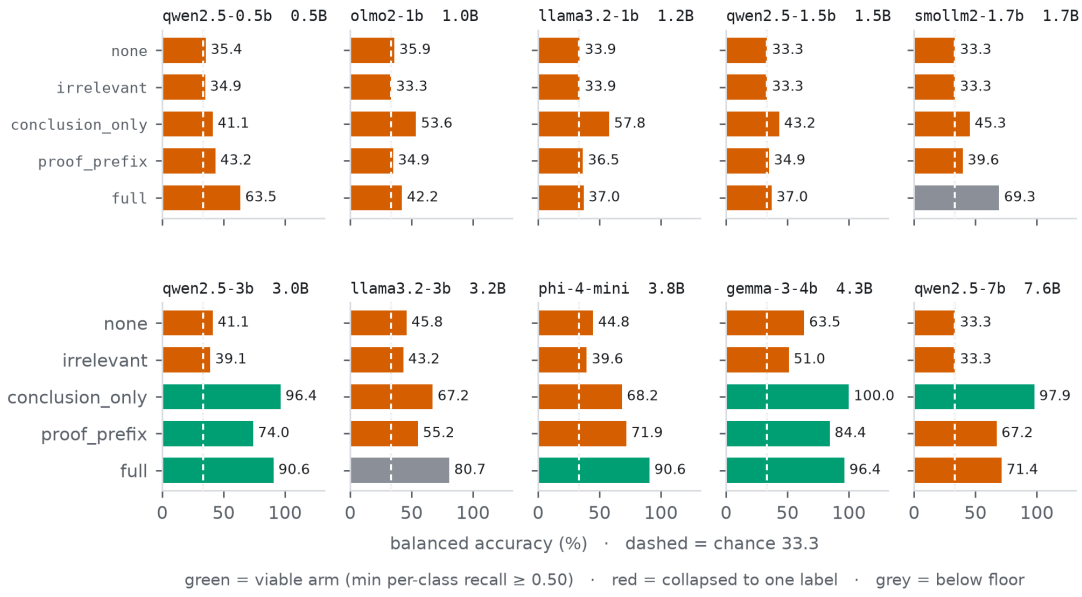


Figure 4: Three-candidate response distribution. Supplying a broken chain lowers the abstention rate from the 75.0% baseline to 42.2% — the opposite of the direction validity detection predicts.



Viability is a property of (model $\times$ arm), not of the model: qwen2.5-7b is viable under conclusion_only (97.9%, min per-class recall 0.938) and not under full (71.4%, 0.141). 4 of 10 models are viable under some arm.

Figure 5: All five arms for all ten models, ordered by parameter count. Chance is 33.3%. Bar colour gives the per-arm viability verdict. Qwen2.5-7B is viable under CONCLUSION_ONLY and not under FULL: viability is a property of (model $\times$ arm).